

Understand Statistical Questions

Lesson 2-1

Name:

Date:

Class:

FRONT OFFICE MISSION

| Scout the Hardwood

You are the new data analyst for the Riverside Hoops youth league. The head coach drops a stack of questions on your desk — some will yield rich data full of variety, others just one fixed fact. Master statistical questions and you decide what the front office actually investigates.



TODAY'S OBJECTIVES

Content Objective: I can tell the difference between a statistical question and a non-statistical question.

Language Objective: I can explain my reasoning using the words statistical question, data, variability, and survey.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Statistical Question

Non-statistical Question

Data

Variability

Survey

Data distribution

Dot plot



Tap any word to see its meaning and a picture.

1

A question that expects a variety of answers and is answered by collecting data is a

2

A question that does not anticipate variability in the data is a

3

Facts or numbers collected to answer a question are

4

How much the values in a data set differ from one another is the

5 A way to collect data by asking many people the same question is a

.

6 The way data values are spread out across a range is the

.

7 A number line that shows one dot for each data value is a

.

Write About the Math

Which question is statistical? Why would the editor's question give the newspaper more interesting data?

¿Qué pregunta es estadística? ¿Por qué la pregunta de la editora le daría al periódico datos más interesantes?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Statistical Question**
Pregunta estadística

☐ **Non-statistical Question**
Pregunta no estadística

☐ **Data**
Datos

☐ **Variability**
Variabilidad

☐ **Survey**
Encuesta

Model: A statistical question expects answers that vary across many cases. — Students identify "How tall is each player?" as statistical because heights vary, while "How tall is our point guard?" has one fixed answer.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The reporter's question is ____ because _____. The editor's question is ____ because _____. The editor's question is better for a story because _____.
- My answer is ____ because _____.
Mi respuesta es ____ porque _____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is _____.
Mi afirmación es _____.
- My evidence is _____, so _____.

Mi evidencia es _____, entonces _____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.